

Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset

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NEMOTRON-CC ABSTRACT

Metrics: MMLU = Massive Multitask Language Understanding; Score = percentage of correctly answered multiple-choice questions (0-100 scale) - **Problem:** Aggressive filtering in FineWeb-Edu, DCLM removes ~90% of data → unsuitable for long-horizon training (15T tokens) - **Solution:** Classifier ensembling + synthetic rephrasing + reduced heuristic filtering - **Results:** - Short horizon (1T tokens, 8B): - High-quality subset: +5.6 pp MMLU vs. DCLM - Long horizon (15T tokens, 8B, 7.2T from Nemotron-CC): - vs. Llama 3.1 8B: +5.0 pp MMLU (70.3 vs. 65.3) - +3.1 pp ARC-Challenge - +0.5 pp average (10 tasks) - Data efficiency (1T tokens): - Full dataset: parity MMLU with DCLM, but 4× more unique real tokens

- **Composition (6.3T total):**

- Deduplicated original tokens: 4.4T
- Synthetic tokens: 1.9T

INTRODUCTION

- Context: Internet crawl = largest source of unique tokens for LLMs
 - Two purposes: high-quality content + diversity

- Status quo (FineWeb-Edu, DCLM):
 - Use model-based filters for high-quality content
 - Boost benchmark accuracies
 - Cost: remove ~90% of data
- Problem - Long-horizon training bottleneck:
 - Recent models train on extended token horizons
 - Llama 3.1 (8B–405B): 15T tokens
 - Gemma 2 27B: 13T tokens
 - Aggressive pruning creates data scarcity for extended training
 - DCLM & FineWeb-Edu contain ~80% near-duplicates (1T and 0.2T unique tokens respectively)
 - Training many epochs on same data = diminishing returns (Muennighoff et al., 2024); better to train on more unique tokens
- Proposed solution: Better accuracy-quantity trade-off via:
 - Classifier ensembling
 - Synthetic data generation
 - Reduced heuristic filter reliance
- Contributions:
 - 6.3T token long-horizon dataset: 4.4T deduplicated original + 1.9T synthetic
 - Released under Common Crawl Terms of Use (CC contributor framework)
 - NeMo Curator library (Apache 2.0): open-source data curation toolkit with reference implementation
 - Quality classifier models released
 - Effectiveness proof vs. DCLM/FineWeb-Edu:
 - 1.1T high-quality subset: +5.6 pp MMLU
 - Full dataset: parity MMLU vs. DCLM, but 4× more unique real tokens
 - 8B @ 15T tokens: +5.0 pp MMLU vs. Llama 3.1 8B (70.3 vs. 65.3)
 - Ablations show:
 - Ensembling selects larger, more diverse high-quality tokens
 - Rephrasing reduces noise + produces fresh variants
 - Disabling heuristic filters boosts high-quality token yield
- Guiding principle: Shift from static heuristic pipeline → learned flywheel
 - Improved data → better LLMs → better synthetic data & quality classification (virtuous cycle)

METHODS

- Overview: Three-component approach to build English Common Crawl pretraining dataset
 - HTML-to-text extraction & heuristic filters (token yield)
 - Model-based quality labeling pipeline
 - Synthetic data generation
- Full pipeline schematic: Figure 3 (Appendix A)

2.1 HTML-to-text Extractor & Filter

- Foundation: Text extraction from HTML is major source of pretraining data
- Reference standard: FineWeb-Edu classifier (model-based quality classifier, proven effective)
- Goal: Balance data quality against token yield, avoid removing good tokens via over-filtering

HTML-to-text Extraction

- Tested: Justext (Pomikálek, 2011) vs. Trafilatura (Barbaresi, 2021)
- Qualitative assessment: both same level
- Quantitative (13 Common Crawl snapshots, Table 1):
 - Justext yields more tokens overall
 - Justext high-quality tokens: +28.6% (by FineWeb-Edu standard: scores 3–5)
 - Strategy: Maximize absolute HQ token count, not percentage
 - Justext: 127B HQ tokens (7.0%)
 - Trafilatura: 104B HQ tokens (7.5%)
- Language filtering: pyltd2 + FastText language identifier (threshold 0.3)
- Deduplication:
 - Global fuzzy deduplication
 - Exact substring deduplication across eighths of snapshots
 - Tools: NeMo Curator library + deduplicate-text-datasets library

Filtering

- Conventional heuristic pipeline (from Parmar et al., 2024):
 - Raffel et al. (2020) + Rae et al. (2021) heuristics
 - KenLM perplexity filter (trained on Wikipedia + books)
- Problem: Heuristic filters remove 18.1% of high-quality tokens (FineWeb-Edu standard)

- Proposal: Apply heuristic filters selectively
 - Do NOT apply to high-quality tokens (identified by model-based classifiers)
 - Apply only to low-quality splits
- Validation: Ablation studies (Section 3.3)

Model-based Quality Labeling

- Problem: Existing classifiers (DCLM, FineWeb-Edu) have limited recall (~10% high-quality tokens)
 - Bottleneck for long-horizon training
 - Quality labels not necessarily aligned with LLM downstream task performance
- Solution: Ensemble-based quality labeling pipeline
 - Build three classifiers with different quality preferences
 - Ensemble scores across all documents
 - Split corpus into quality buckets
 - Regroup into 5 quality levels based on downstream task performance

Quality Classifier Training

- Two annotation versions: prompt Mistral 8x22B-instruct + Nemotron-340B-instruct
 - Score documents from FineWeb on educational value (0–5 scale)
- Fine-tune linear regression models on Snowflake-arctic-embed-m embeddings (text-to-vector conversion model)
 - Frozen embedding + encoder layers (weights unchanged during training, only linear layer updated)
 - 20 epochs, learning rate 0.0003
 - Checkpoint selected by highest F1 on validation set
- Third classifier: DCLM classifier (fastText-based, Li et al., 2024)
 - Trained on instruction-formatted data + ELI5 subreddit high-scoring posts (Explain-Like-I'm-5: community-curated, high-quality explanations)
 - Stronger performance identifying high-quality tokens vs. FineWeb-Edu
 - Offers different perspective for increased recall

Quality Scoring and Bucketing

- Each classifier predicts quality scores for all documents

- Round scores to integers 0–19 (20 equal-sized buckets, ~5% of documents per bucket)
 - Bucket 19 = top 5% highest quality
- Final quality score: ensemble via maximum operation across three classifiers
 - Document distribution across buckets skewed by ensemble operation

Quality Labeling

- Map fine-grained scores (0–19) to 5 downstream quality categories
- Method: continuous pretraining annealing
 - Setup: 50B token training on 70% pretrained 8B model
 - Metric: average performance across 9 tasks per bucket
 - Weighted mix: 66% standard baseline data + 34% evaluated bucket
 - (Tests each bucket's contribution; higher gain = higher quality category)
- Group 20 buckets into 5 quality categories based on performance

Synthetic Data Generation

- Observation: Low-quality docs contain noise/errors; high-quality have good writing/formatting
- Strategy: Different approaches for low vs. high quality

Low-Quality Data

- Goal: Reduce noise/errors while preserving useful information (cut training compute)
- Method: Rephrasing using medium-sized LM (Maini et al., 2024)
 - NOT creating new knowledge (unlike textbook/story generation approaches)
 - Transform text to Wikipedia style (Prompt 5, Appendix H)
 - Reduces errors, redundancies, improves formatting
- Result: Enhanced parallel corpus, lower perplexity, better downstream performance

High-Quality Data

- Goal: Generate unique tokens, condense essential knowledge
 - Rationale: Repeated tokens show diminishing returns after 4 epochs (Muennighoff et al., 2024)

- Four synthesis prompts:
 - Diverse Question-Answer pairs: ask varied question types (yes/no, open-ended, multiple-choice) with correct answers
 - Distill: rewrite as concise, clear passage
 - Extract knowledge: rewrite only informative content, discard filler
 - Knowledge list: structure key information as organized list
- Requirements: Clear, concise responses preserving factual info + concrete details (numbers, etc.)

Processing

- Chunking: Split documents into segments (complete lines, token limit enforced)
 - Prevents over-simplified model outputs on long inputs
 - Discard over-length lines exceeding token limit
- Post-processing:
 - Remove incomplete results, markdown formatting (e.g., **bold**)
 - Strip prefixes (“Here is a paraphrased version:”, “Paraphrased Text:”)
 - Remove enclosing quotation marks
 - Filter under-length outputs (< 50 tokens)
 - Wikipedia results: concatenate passages from same original document
 - Diverse QA results: shuffle pairs, append to segment end

Scale & Implementation

- Model: Mistral NeMo 12B (instruct, FP8 inference)
 - top-p: 0.9, temperature: 0.5
- Synthesis (Table 3):
 - Low-quality (403.0B raw) → Wikipedia → 336.3B synthetic
 - High-quality (451.3B raw):
 - Wikipedia: 372.9B
 - Diverse QA Pairs: 499.5B
 - Distill: 157.6B
 - Extract Knowledge: 303.6B
 - Knowledge List: 203.2B
 - Total synthetic: 1.8T tokens
 - Medium-quality: not synthesized (resource constraints)

- Tools: TensorRT-LLM + NeMo-Skills

Putting It All Together

- Combine all techniques above on 99 Common Crawl snapshots (CC-MAIN-2013-20 through CC-MAIN-2024-30)
- **Nemotron-CC dataset: 6.3T tokens**
 - 4.4T globally deduplicated original tokens
 - 1.9T synthetically derived tokens
- Comparison to baselines:
 - ~4× more unique tokens vs. FineWeb-Edu-2 and DCLM
 - Reason: DCLM/FineWeb used sharded approximate deduplication only, contain ~80% fuzzy duplicates
- High-quality subset (for fairer short-horizon comparison):
 - Nemotron-CC-HQ: 1.1T tokens
 - Composition: highest-scoring real data + diverse QA pair synthetic variants only

Dataset	Total	Unique	Synthetic
FineWebEdu-2	5.4	1.1	-
FineWebEdu	1.3	0.2	-
DCLM	3.8	1.0	-
Nemotron-CC	6.3	4.4	1.9
Nemotron-CC-HQ	1.1	0.6	0.5

Dataset sizes in trillions of tokens. “Unique” shows the estimated number of tokens after global fuzzy deduplication of the real tokens.